The best plan is at a corner

Optimization · Level 2 · Lesson 2



Today we will

- Draw each limit of the garden as a line and shade the plans that fit.
- Find the best plan at a corner by sliding the profit line.
- See how a price or a limit change moves the best corner.

Remember

- Each limit is a line; the plans that keep every limit are the shaded shape.
- The best plan of a linear program is always at a corner.
- Binding limits are used up at the best corner; spare ones don’t matter there.

The garden on graph paper

1. Warm-up (5 min): Tomatoes or beans: what would you plant?
2. Teach (8 min): Lines, the shaded shape, the corners.
3. Draw (12 min): Groups draw the three limits and shade.
4. Corners (10 min): Profit at every corner; circle the best.
5. What if (5 min): Tomatoes at 50: does the corner move?

Let 1 square be 1 bed, from 0 to 18 across and 0 to 14 up, so every line fits. No graph paper? Rule a grid on plain paper or draw one big graph on the board.

Let’s discuss

1. Why is the best plan at a corner and not in the middle?
2. Why doesn’t extra water change the best plan?
3. Should profit be the garden’s only goal? What else matters?

Exit ticket

- Sketch the shaded shape and mark the best corner. Which two limits meet there?